

Bail(ey)ing Out

Bailey Hall is one of the largest public spaces at Cornell University. Used for academic events, concerts, and convocations, this stately building is shaped as an amphitheatre: a semicircular auditorium, with a corridor and colonnaded portico wrapping around it, and a main entrance through a foyer on the south side. As with any building on campus, the number of people allowed in the Bailey Hall auditorium is limited not only by acoustics and convenience, but also by the NY State Fire Code – to ensure a quick and orderly exit, particularly in case of emergency. While the University already has an excellent evacuation and safety plan in place, ***your team's goal is to explore whether leaving this building can be made both easier and faster*** without major changes to its infrastructure and appearance. To do this, you will ideally consider both emergency evacuations and ordinary departures (e.g., the audience flowing out after a large lecture or concert).

Crowd motion can be complicated even without any emergency. People make their individual choices (*which exit to go to? along which path? how quickly? what to do if there is a traffic jam or the preferred route is somehow blocked?*). They make those choices individually, but clearly affect each other in the process. Some of them might be brutally rational (*what's the quickest way out?*), others may be influenced by habit (*how did I enter this room?*), peer pressure (*which way is everyone going?*), or social norms (*it would be rude to rush ahead of others!*). The majority operate with only partial knowledge of their environment (*which doors are really open right now? how many people will choose each of them?*). What appears to be the best strategy for each individual might end up impeding the motion of the whole group. Traffic engineers know that, when it comes to independent decision makers, restricting the range of choices they have can sometimes improve the outcomes for everyone. This is particularly important in evacuation scenarios, where Bailey Hall would have to be emptied in under 180

seconds – from the moment when the emergency is declared and until the last person leaves the building. For large non-academic events (e.g., concerts or performances involving more than 500 people), the state law requires the presence of at least one "Crowd Manager" for every 250 attendees. Any such event that fills all 1,329 seats in Bailey Hall is thus required to include at least six Crowd Managers – specially trained ushers, who can provide detailed evacuation instructions to the audience. However, guiding and assisting that many people is difficult, particularly if this is not a drill but a real emergency. Can we make their jobs easier through minor changes to the environment?

Develop and use a mathematical model to address (some of) the following questions:

1) How long does it take for the audience to leave Bailey Hall after a concert?

2) How long would it take for the same concert audience to leave in case of emergency while guided by Crowd Managers? How would that egress time change in case of an unexpected (and possibly initially unknown) obstruction of one of the exits?

3) How would the above differ from the time needed to evacuate the same number of students from a large lecture taught in Bailey Hall? Note that Crowd Managers are not present during academic events since students are very familiar with their regular lecture buildings. They also participate in fire drills and are assumed to know the location of emergency exits.

4) Could these building egress times be decreased safely
    a) by using temporary/removable barriers in the corridor to restrict people's paths from the mezzanine exits to specific exterior doorways?
    b) or by attaching directional LED light strips to the floors or walls to guide different parts of the audience to specific exits?

    (For concerts, both of these measures are meant to help Crowd Managers rather than replace them. Their minimum number is mandated by the state law.)

5) Would such measures be cost-effective? E.g., how would their installation and maintenance costs compare with the cost of having more Crowd Managers on duty?

While working on these questions, you should keep in mind that good evacuation guidance should be robust to execution error. A good plan must balance efficiency and simplicity since an erroneous implementation by confused/scared people might easily increase the egress times.

In addition to your technical description of the model and the results, write a one-page executive summary of your findings addressed to Mr. Timothy Ryan, University Fire Marshal and Director of Fire Protection and Emergency Services.

A word of advice from CMCM organizers:

We provide fairly detailed information about Bailey Hall below. But you might want to start with a simplified schematic version of this building, retaining the main features of the problem only. You will need to balance the level of realism in your representation of the building against the time required to explore people's logic as they make different choices on their way out. Trying to tackle both of these aspects in detail would be very difficult in a single weekend. You can (and probably should!) also focus on a subset of our questions instead of doing a little bit on each of them. The organizers view Questions (2), (3), and (4) as the most important. We encourage you to complete at least one of Questions (2) & (3), and then use the results as your baseline for Question (4). In the latter, you can evaluate either one or both suggested interventions. Both Questions (1) and (5) should be treated as optional extensions.

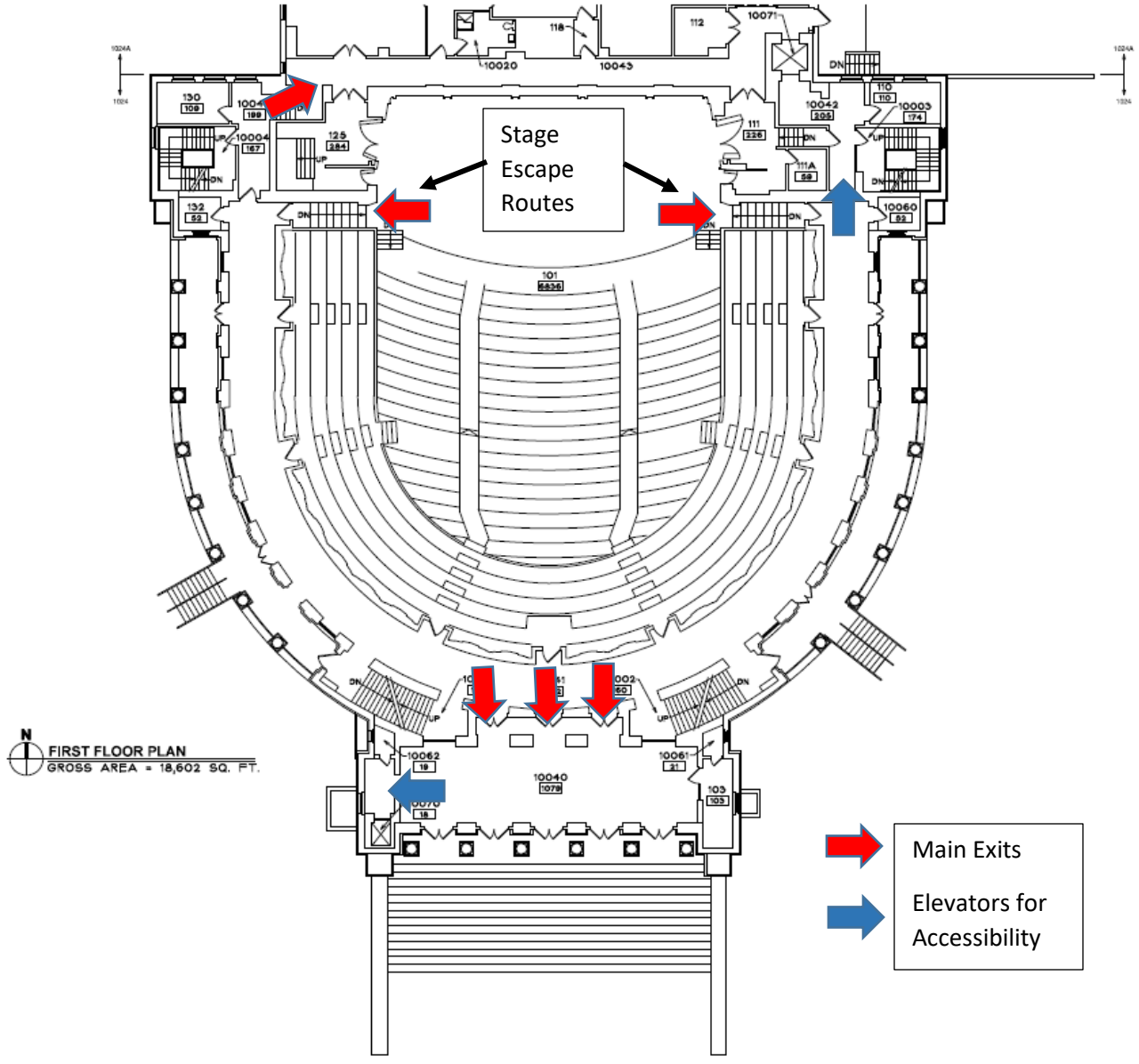
Whichever questions you choose, please state them explicitly in your manuscript, and make sure to include the executive summary with your conclusions. Start simple; add bells and whistles later. And, most importantly, don't forget to have fun in the process!

Building details:

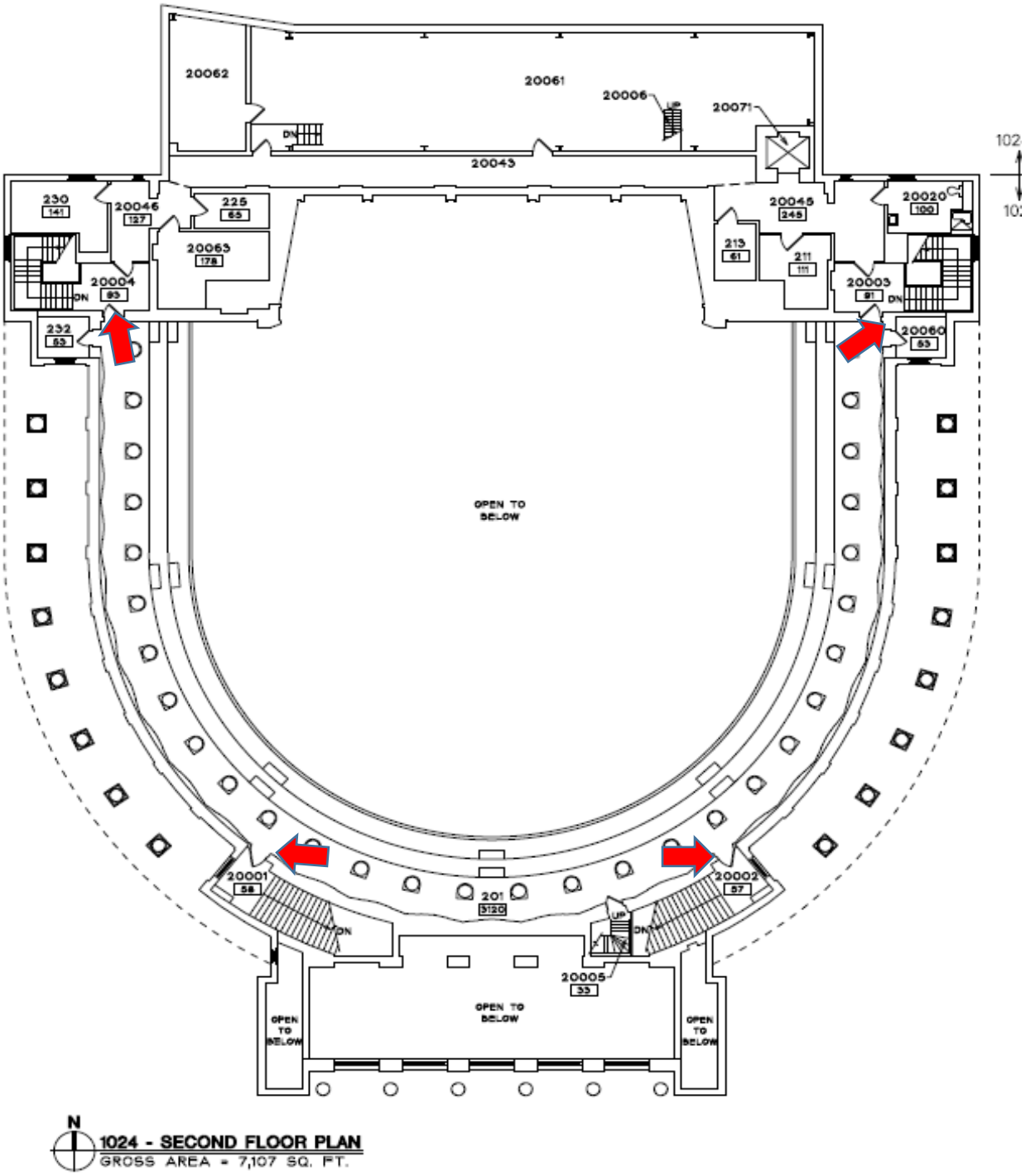
The Bailey Hall auditorium has two floors, with the main space on the first floor that includes an orchestra (527 seats) and an elevated mezzanine/loge level (486 seats). The balcony is on the second floor and includes 316 additional seats. (These numbers include up to 3 wheelchairs, which are not included in the seat count on the attached seating maps.) The mezzanine is separated from the orchestra by a barrier, which also affects the exit paths and crowd dynamics. People from the orchestra use six access points to ascend to the mezzanine. All nine regular exit doors[1] from this first level of the auditorium are through the mezzanine, spread out along the semicircle. These nine doors lead to a semicircular corridor, which has four (rarely used) side doors to the portico and three doors to the foyer. (See the attached evacuation plans.) The balcony is connected to the first floor by two staircases near the foyer (each with two flights of stairs and barriers separating people into two lanes) and staircases on each side of the stage leading directly outside (also rarely used except in emergency scenarios). The usual first floor exit is currently through the five large doors from the foyer, leading to a monumental staircase outside. Two narrower staircases (one on each side of the building) also lead from the portico to the street level. In addition, there is also an ADA-compliant exit using an elevator on the west side of the foyer, but its use in evacuation scenarios is prohibited.

[1] Bailey Hall has several additional service/performer spaces behind and on either side of the stage, with special exits from them. It also has an entire ground level, which in reality would also have to be evacuated. For simplicity, we ask you to ignore these additional details.

Main Auditorium – First Floor



Main Auditorium – Balcony

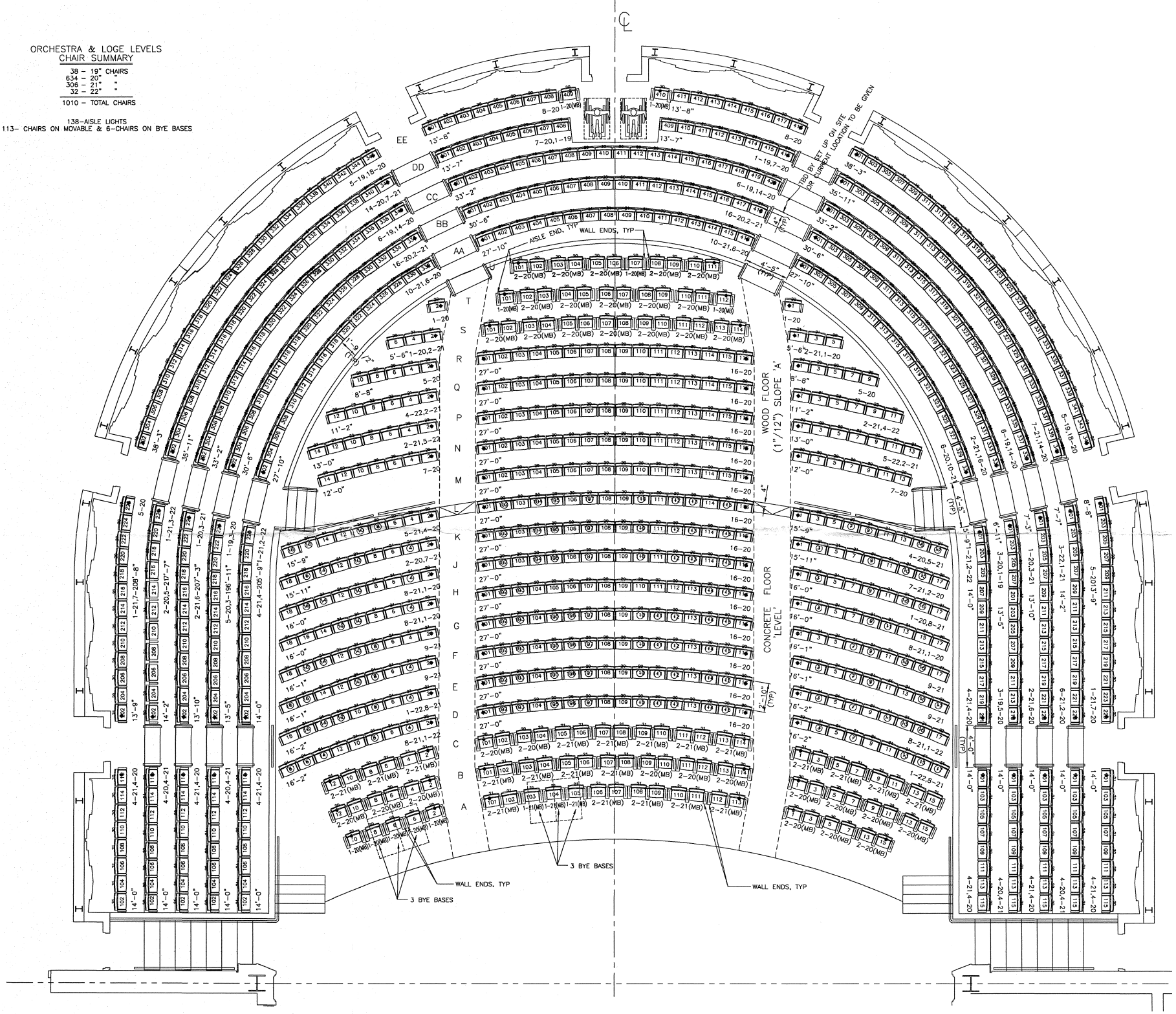


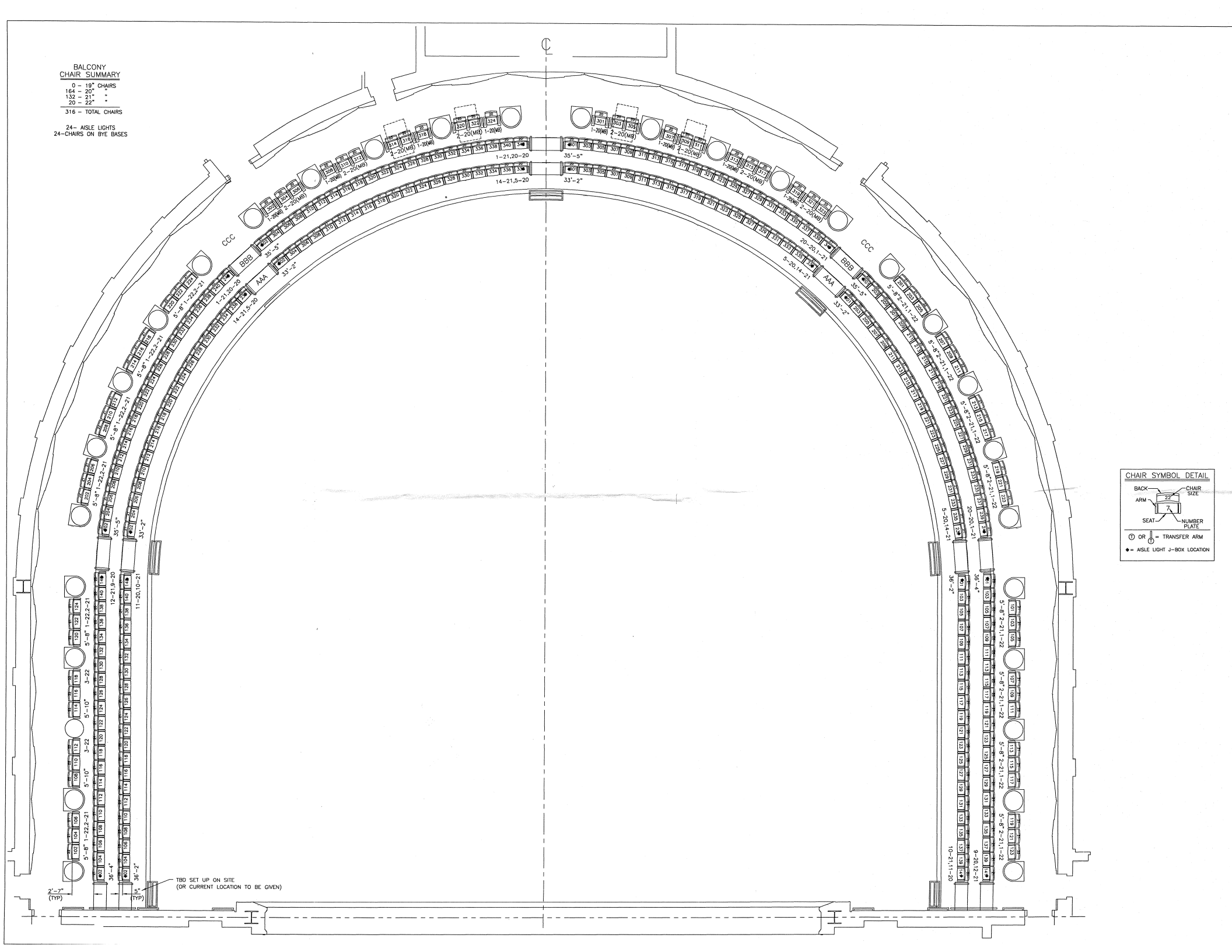
ORCHESTRA & LOGE LEVELS
CHAIR SUMMARY

38 – 19" CHAIRS
634 – 20" "
306 – 21" "
32 – 22" "

1010 – TOTAL CHAIRS

138–AISLE LIGHTS
113– CHAIRS ON MOVABLE & 6–CHAIRS ON BYE BASES




BALCONY
CHAIR SUMMARY
0 – 19" CHAIRS
164 – 20" "
132 – 21" "
20 – 22" "
316 – TOTAL CHAIRS
24– AISLE LIGHTS
24–CHAIRS ON BYE BASES
CCC
BBB
AAA
1–21,20–20
14–21,5–20
1–20(MB)
2–20(MB)
35'–5"
33'–2"
5'–8" 1–22,2–21
5'–8" 2–21,1–22
20–20,1–21
5–20,14–21
12–21,9–20
11–20,10–21
10–21,11–20
9–20,12–21
3–22
5'–10"
36'–4"
36'–2"
2'–7" (TYP)
5" (TYP)
TBD SET UP ON SITE
(OR CURRENT LOCATION TO BE GIVEN)
CHAIR SYMBOL DETAIL
BACK
CHAIR SIZE
ARM
SEAT
NUMBER PLATE
OR = TRANSFER ARM
= AISLE LIGHT J–BOX LOCATION